Ans: C

Q6

For the mass of 500 g to move downwards and the mass of 200 g to move upwards, the friction F in the pulley must act opposite to the motion of the masses.

For the masses to move at constant speed, net force = 0. Considering the pulley and masses as a whole, $Mg - F - mg = 0$ or $F = (M - m)g$

$F = (0.500 - 0.200) \times 9.81 = 2.94$ N

Speed of rope = $v = r\omega = 0.035 \times 2.4 = 0.084$ m s^{-1}

Hence, rate of work done = $P = Fv = 2.94 \times 0.084 = 0.25$ W

Since the force exerted by the motor on the rope is opposite in direction to the velocity of the rope, the rate of work done = - 0.25 W

Ans: B

Q7